SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program/s: BTECH — Year: 2nd Semester: IV

Stream/s : CSE (Cybersecurity)

Subject: Introduction to Cryptography — Time: 03 hrs (10:00 to 01:00 PM) AM

Date: 30 / 06 / 2023 — No. of Pages: 04

Marks: 100

Re- Examination (2021-22/2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Alphabet Codings

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Inverses in Z_{26}

a	1	3	5	7	11	17	25
a^{-1}	1	9	21	15	19	23	25

Q1		Answer briefly:	Marks
Q1 CO-1 BL-3	a.	The following text has been obtained by first encoding the plaintext by using Vigenére Cipher with the key "ARENA" and then encoding the result again by Permutation Cipher with the key. $\pi = \begin{array}{\|c\|c\|c\|c\|c\|} \hline 1 & 2 & 3 & 4 & 5 \\ \hline 5 & 4 & 2 & 3 & 1 \\ \hline \end{array}$	[05]

		Decrypt the text: HGFVNRRZZR	
Q1 CO-1 BL-3	b.	Suppose that the first seven bits of the key stream of a LFSR of degree 3 is 10100111. Find next three bits of this key stream.	[05]
Q1 CO-1 BL-3	c.	Decrypt the following ciphertext obtained from the Autokey Cipher by using the key $k = 12$: OJHMB	[05]
Q1 CO-3 BL-3	d.	Define a hash function $h : \mathbb{Z}_{11^2} \to \mathbb{Z}_{23}$ by $h(m) \equiv 5^a 7^b$ mod 23 where $m = a + 11b$ with $0 \leq a, b \leq 10$. Compute h (70).	[05]
Q2 CO-3 BL-3	a.	Suppose that $f: \{0, 1\}^m \to \{0, 1\}^m$ is a preimage resistant bijection. Define $h : \{0, 1\}^{2m} \to \{0, 1\}^m$ as follows: Given $x \in \{0, 1\}^{2m}$, write $x = x' \| x''$ where $x', x'' \in \{0, 1\}^m$. Then define $h(x) = f(x' \oplus x'')$. If $x_0 = 00 \cdots 0000 \cdots 00 \in \{0, 1\}^{2m}$ and $x_1 = 00 \cdots 0100 \cdots 01 \in \{0, 1\}^{2m}$, compute $h(x_0)$ and $h(x_1)$ in terms of f and state whether h is a second preimage resistant.	[10]
Q2 CO-1 BL-3	b.	Solve $12x + 4 = 10$ mod 26 and find both the solutions by using the fact that $6(11) = 66 = 5(13) + 1$	[10]
Q3 CO-2 BL-3	a.	Bob accidentally selected n such that $n + 9^2$ is a perfect square, that is, $n + 9^2 = s^2$ for some positive integer s. Find the factors of n. Hint: $a^2 - b^2 = (a + b)(a - b)$.	[10]
Q3 CO-2 BL-3	b.	Using the result of previous problem, find the factors of $n = 2189284635403183$.	[10]

<table>
<tr>
<td>Q4
CO-2
BL-3</td>
<td>a.</td>
<td>
Using square-and-multiply algorithm:

Square-and-Multiply Algorithm (x, b, n):

- Step 1: $z \leftarrow 1$
- Step 2: For $i = f - 1$ down to 0, do
  - Step 2.1: $z \leftarrow z^2 \pmod{n}$
  - Step 2.2: If $b_i = 1$, then $z \leftarrow z \times x \pmod{n}$
- Step 3: Return z

compute $10^9 \bmod 17$ by filling out the following table, where b_i 's are binary bits of 9.

Note: No marks will be awarded if the table is left blank and no marks will be awarded for directly computing $10^9 \bmod 17$ without using square-and-multiply algorithm.

<table>
<tr><td>i</td><td>b_i</td><td></td><td>z</td></tr>
<tr><td></td><td></td><td></td><td>1</td></tr>
<tr><td>3</td><td></td><td></td><td></td></tr>
<tr><td>2</td><td></td><td></td><td></td></tr>
<tr><td>1</td><td></td><td></td><td></td></tr>
<tr><td>0</td><td></td><td></td><td></td></tr>
</table>
</td>
<td>[10]</td>
</tr>
<tr>
<td>Q4
CO-2
BL-3</td>
<td>b.</td>
<td>Let M be the Multiplicative Cipher where the encryption rule is given by $e_k(x) = kx$ mod 26 and S be the Shift Cipher. Consider the key (5,14) in M x S. Find the corresponding key in S x M that gives the same encryption rule.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q5
CO-2
BL-3</td>
<td>a.</td>
<td>To use ElGamal cryptosystem, Alice selects the prime number p = 29 and g = 2. She selects private key a = 10 and computes A. Find the value of A. Remember that answer should be an integer between 0 and 29.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q5
CO-2
BL-3</td>
<td>b.</td>
<td>Upon receiving A from Alice as in the previous problem, Bob selects his secret random element k = 8 to send the message m = 6. He computes the ciphertext (y_1, y_2). Compute the value of y_1 and y_2.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q6
CO-1
BL-3</td>
<td>a.</td>
<td>Suppose that the Affine Cipher with the encryption function $e(x) = 3x + 5 \bmod 26$ is used to encrypt a plaintext. Find the decryption function: $d(y) = cy + d \bmod 26$.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q6
CO-1
BL-3</td>
<td>b.</td>
<td>If the Affine Cipher in the previous problem yields the following ciphertext, find the corresponding plaintext: XVMO</td>
<td>[10]</td>
</tr>
</table>

Q7 CO-2 BL-3

a.

Consider the following S-box π_S for a simple SPN:

input	000	001	010	011	100	101	110	111
output	110	101	001	000	011	010	111	100

Compute $N_L(6, 4)$ by filling out the following table. Also compute the bias ϵ (6,4)

X_1	X_2	X_3	Y_1	Y_2	Y_3	6	4
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Given reason the pair (6,4) can/cannot be used to construct a linear approximation for the given SPN.

[10]

Q7 CO-1 BL-3

b.

Find the inverse of the following matrix in Z_{26}.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

[10]